

Design and application of a decatungstate-based ionic liquid photocatalyst for sustainable hydrogen atom transfer reactions

Miguel Claros,^a Julian Quévareq,^a Sara Fernández-García,^{a,b} Timothy Noël^a

^a Flow Chemistry Group, Van't Hoff Institute for Molecular Sciences (HIMS), Universiteit van Amsterdam (UvA), 1098 XH, The Netherlands.

^b Facultad de Química, Universidad de Murcia, Centro Multidisciplinar Pleiades-Vitalis, Campus de Espinardo, 30100 Murcia, Spain.

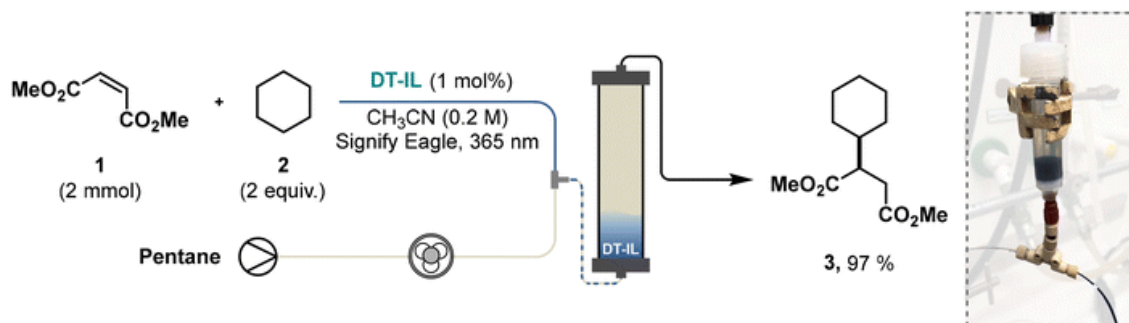
e-mail: m.claroscasielles@uva.nl

Keywords: Photochemistry • Hydrogen-Atom-Transfer • Ionic-Liquid • Green-Chemistry

Photocatalysis offers a sustainable alternative to thermally driven synthesis, yet most photocatalytic transformations rely on homogeneous catalysts that are difficult to recover and recycle.^[1,2] Decatungstate is a highly robust hydrogen atom transfer (HAT) photocatalyst that exhibits exceptional photostability, making it an attractive candidate for sustainable large-scale applications.^[3]

To enhance its tunability and facilitate catalyst recovery, we envisioned the development of decatungstate-based ionic liquids (DT-ILs), combining the photocatalytic activity of decatungstate with the advantageous properties of ionic liquids, including improved solvent compatibility, thermal stability, and separability. Despite previous reports describing DT-ILs in thermal oxidation chemistry, their application as recyclable photocatalysts has remained unexplored.

Herein, we report the synthesis and characterization of a phosphonium-based DT-IL and investigate its performance in a range of photocatalytic HAT transformations under both batch and continuous-flow conditions. The catalyst displays excellent activity, broad applicability, and straightforward recyclability, highlighting the potential of DT-ILs as sustainable photocatalytic platforms for scalable organic synthesis.



References

- [1] T. Noël and E. Zysman-Colman. *Chem. Catal.*, **2021**, 1, 403.
- [2] A. Chaudhuri, S. D. A. Zondag, J. H. A. Schuurmans, J. van der Schaaf and T. Noël. *Org. Process Res. Dev.*, **2022**, 26, 1279.
- [3] Z. Wen, D. Pintossi, M. Nuño and T. Noël, *Nat. Commun.*, **2022**, 13, 6147.
- [4] Y. Martinetto, B. Pégot, C. Roch-Marchal, M. Haouas, B. Cottyn-Boitte, F. Camerel, J. Jętic, D. Morineau, E. Magnier and S. Floquet, *New J. Chem.*, **2021**, 45, 9751.
- [5] M. Claros, J. Quévareq, S. Fernández-García, T. Noël. *Green Chem.*, **2025**, 27, 7660.